Module 1: Machine Setup and Control Familiarization

AR System Guidance

- Display a reminder to read the hazards, PPE, and Key Terms and put on **approved safety glasses** before walking up to the mill. Require operator acknowledgment before presenting the machine controls.
- Highlight the **toolbox** and identify the **collet**, **covered end mill**, **drawbar wrench**, **vise handle**, **parallels**, **wax blocks**, **shop rag**, and **chip brush** one at a time. Display Figure 1 and the matching source definitions so the operator can identify each item.
- Highlight and label the two **X-axis handwheels** at the **table** ends, the front **Y-axis handwheel** below the **table**, and the front-left **Z-axis crank**. Use Figure 2 to distinguish their locations.
- Highlight the **X-axis handwheel at the right end of the table**. Display the source directions from Figure 3: clockwise moves the **table** left; counterclockwise moves it right. Keep the control identity visible so these directions are not applied to the left-end **handwheel**.
- Highlight the **Y-axis handwheel** and display the directions from Figure 4: clockwise moves the **table** in, away from the operator; counterclockwise moves it out, toward the operator.
- Highlight the **Z-axis crank** and display the directions from Figure 5: clockwise raises the **table**; counterclockwise lowers it.
- Prompt the operator to try each control slowly, pushing the **handwheel or crank** inward while turning it to keep it engaged. Display a reminder to stand to the side of the **handle** and keep the other hand off the **table**. Prompt observation of actual **table** motion; do not animate an unspecified amount of travel.
- Highlight the **main power switch** on the **gray electrical box** and display Figure 6 showing OFF. Prompt the operator to confirm the switch is OFF before **tool** installation. If it is not OFF, pause the AR workflow until the source-required condition is satisfied (for vision model).
- Allow the next module only after **tool** identification, control familiarization, and the main-power OFF condition have been confirmed.

[REFERENCE IMAGES: Figures 1 to 6 show the tool layout, axes, rotation directions, and main-power OFF position.]

Operator Procedure

1. Read the hazards, required PPE (personal protective equipment), and Key Terms in this SOP. Put on **approved safety glasses** (if not using AR glasses), Tie back long hair, long beards, wear closed-toed shoes, take off watches and any jewelry around wrists and necks, tuck in your shirt, roll up any long sleeves, and make sure that you don’t have any baggy clothes on before walking up to the mill. (make list into bullet points)

2. Find the tools on top of the **toolbox** next to the mill: the **collet**, **end mill** with its **protective cover**, **drawbar wrench**, **vise handle**, **parallels**, **wax blocks**, **shop rag**, and **chip brush** (Figure 1). Each is explained in the Key Terms.

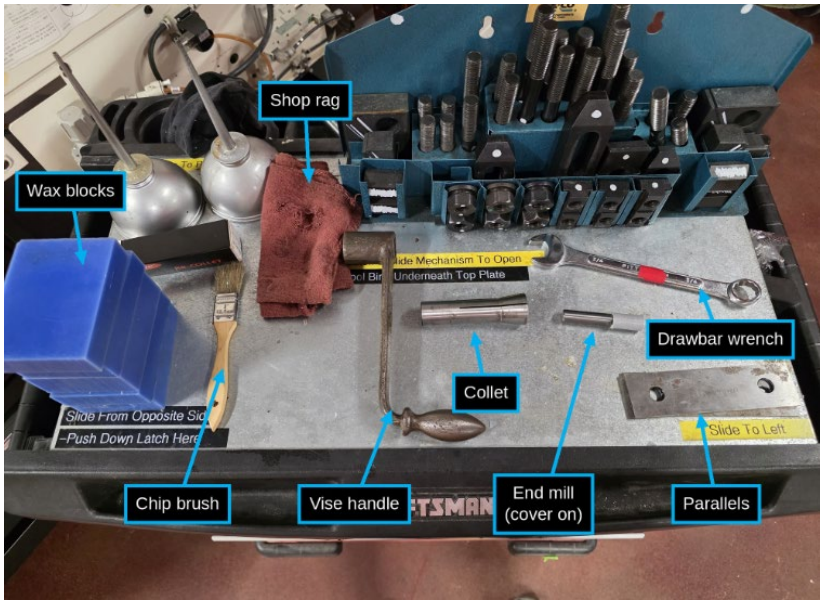


Figure 1. Tools on top of the toolbox next to the mill.

3. Find the **handwheels and crank** that move the **table** (Figure 2). The **X-axis handwheels** are at the left and right ends of the **table** and move the **table** left and right. The **Y-axis handwheel** is on the front of the machine, just below the **table**, and moves the **table** in and out. The **Z-axis crank** is the long **crank handle** on the front-left of the machine, below the **table**, and moves the **table** up and down.

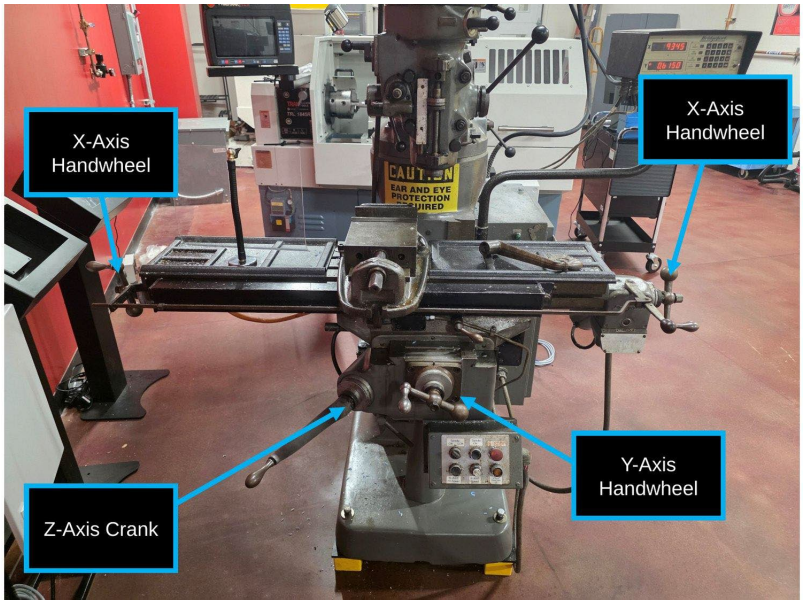


Figure 2. The handwheels and crank that move the table. There is an X-axis handwheel at each end of the table.

4. Try each one. Push the **handwheel or crank** inward toward the machine while you turn it, so it stays engaged and does not come loose. Turn it slowly and watch which way the **table** moves (Figures 3 to 5). Stand to the side of the **handle** you are turning, and keep your other hand off the **table**.

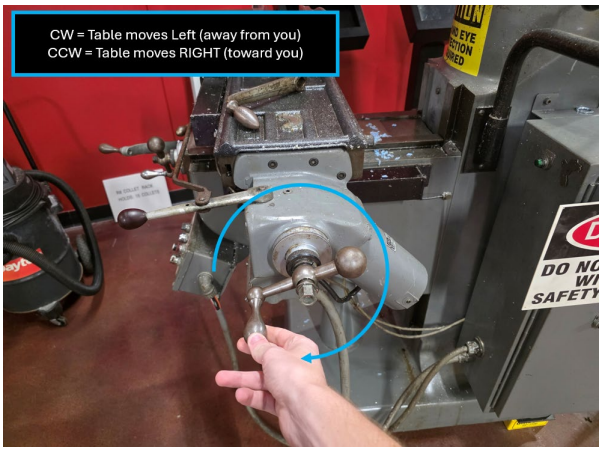


Figure 3. X-axis handwheel at the right end of the table: turning it clockwise moves the table to the left; counterclockwise moves it to the right.

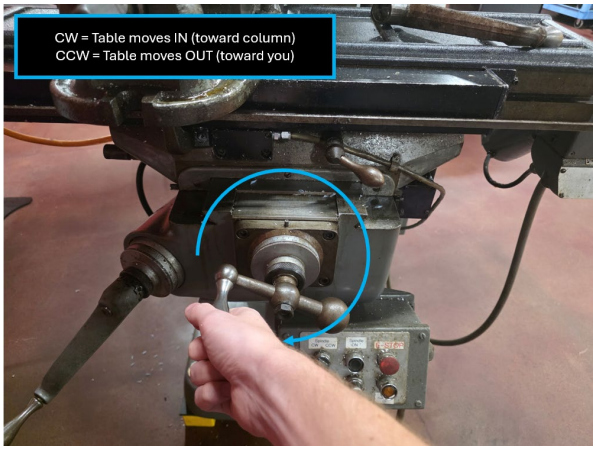


Figure 4. Y-axis handwheel: turning it clockwise moves the table in, away from you; counterclockwise moves it out, toward you.

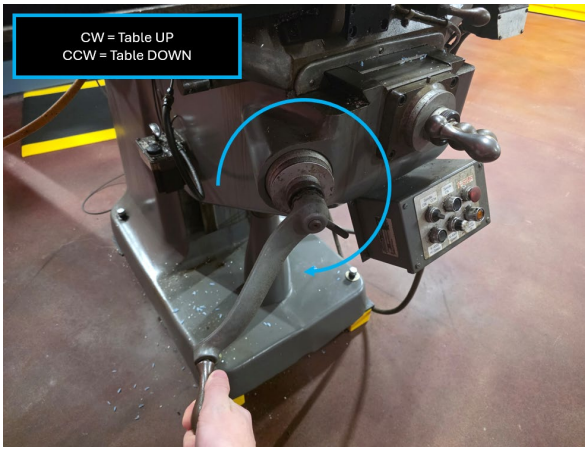


Figure 5. Z-axis crank: turning it clockwise raises the table; counterclockwise lowers it.

5. Find the **main power switch** on the **gray electrical box** next to the mill and make sure it is turned to OFF (Figure 6). With the power off, the **spindle** cannot start while your hands are near the **cutting tool** during the next steps.



Figure 6. Main power switch on the gray electrical box, turned to OFF.

Module 2: Installing the Collet and End Mill

AR System Guidance

- Highlight the **main power switch** and require a fresh operator confirmation that it is OFF. Highlight the **spindle** opening shown in Figure 11. Do not display **tool**-insertion prompts before this check.
- Highlight the **vise** if it is directly under the **spindle**. Prompt the operator to use the **X-axis handwheel** to move the **vise** out of the way and ask the instructor to check before proceeding. Do not invent a required travel distance or rotation direction for this repositioning.
- Highlight the **end mill and collet** on the **toolbox** using Figure 1. Label the **smooth shank** and **sharp flutes** using Figures 7 and 8. Prompt the operator to pick up the **tool** by its **smooth end** and leave the **protective cover** on the cutting end.
- Highlight the narrow, slotted end of the **collet**. Show an insertion arrow along the **tool shank** into that end and display Figures 9 and 10. Prompt the operator to insert the **smooth shank** until only the covered cutting end protrudes.
- Highlight the **spindle** opening and display Figure 11. Show an upward insertion arrow for the assembled **collet** and **covered end mill**. Prompt the operator to turn the **collet** slightly until it seats fully; do not specify a rotation direction for this slight seating motion.
- Highlight the **drawbar** on top of the machine. Display Figure 12 and a clockwise arrow for hand tightening. Prompt the operator to keep supporting the **collet** from below while turning the **drawbar** until snug.
- Display a reach reminder before any **drawbar** or brake action: stay on the floor and ask the instructor for help if the control cannot be reached. **Block** the AR step if the operator reports being unable to reach safely; do not suggest standing on an object.
- Highlight the **spindle brake lever** using Figure 13 and prompt the operator to hold it firmly so the **spindle** cannot turn. Display the source reminder that more force than expected may be required.
- Highlight the **drawbar wrench** and **drawbar**, display Figure 14, and show the source clockwise tightening direction. Prompt the operator to tighten until snug while holding the brake. If the operator cannot hold the brake and reach the **drawbar** with the **wrench** simultaneously, direct them to ask the instructor to assist with one of those actions.
- Prompt the operator to release the **spindle brake**, remove the **wrench** from the **drawbar**, and return the **wrench** to the **toolbox**. Require confirmation that the **wrench** has been removed before continuing.
- Display Figure 15 and highlight the **protective cover** as the hand-contact location. Prompt the operator to pull down gently on the **covered end mill** and confirm that it does not move or drop. Do not claim that computer vision can verify clamping force.
- Do not advance until the operator confirms that the **collet** is seated, the **drawbar** is snug, the **wrench** is removed, the brake is released, the **protective cover** remains on, and the **end mill** does not move during the source check. If the **tool** moves or drops, pause the workflow and direct the operator to obtain instructor assistance.

[REFERENCE IMAGES: Figure 1 and Figures 6 to 15 identify the tools, covered tool handling, collet assembly, spindle insertion, drawbar, brake, and retention check.]

Operator Procedure

1. Find the **spindle**: the round opening under the upper part of the machine where the **cutting tool** goes (Figure 11).

2. If the **vise** is directly under the **spindle**, turn the **X-axis handwheel** to move the **table** until the **vise** is out of the way. Ask your instructor to check before you continue.

3. Go to the **toolbox** and find the **end mill** and the **collet** (Figure 1).

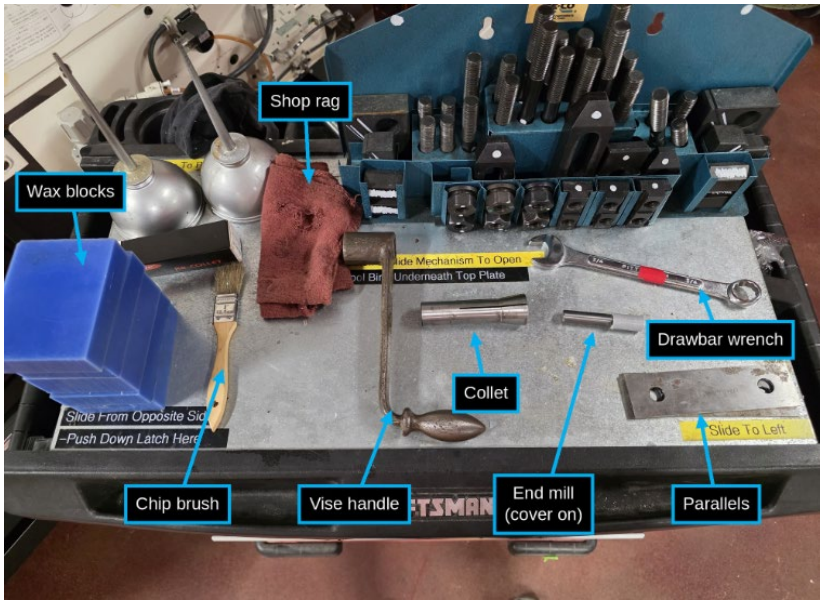


Figure 1. Tools on top of the toolbox next to the mill.

4. Pick up the **<u>end mill</u>** by its **<u>smooth end (shank)</u>**. The **protective cover** should be on the sharp end; leave it on (Figure 7). Never grab the **sharp spiral cutting edges (flutes)** (Figure 8).

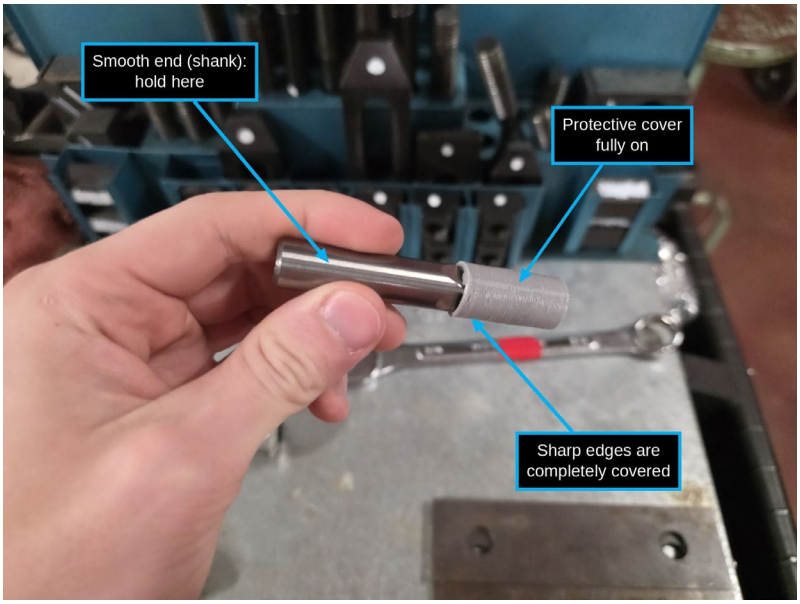


Figure 7. End mill with its protective cover fully on.

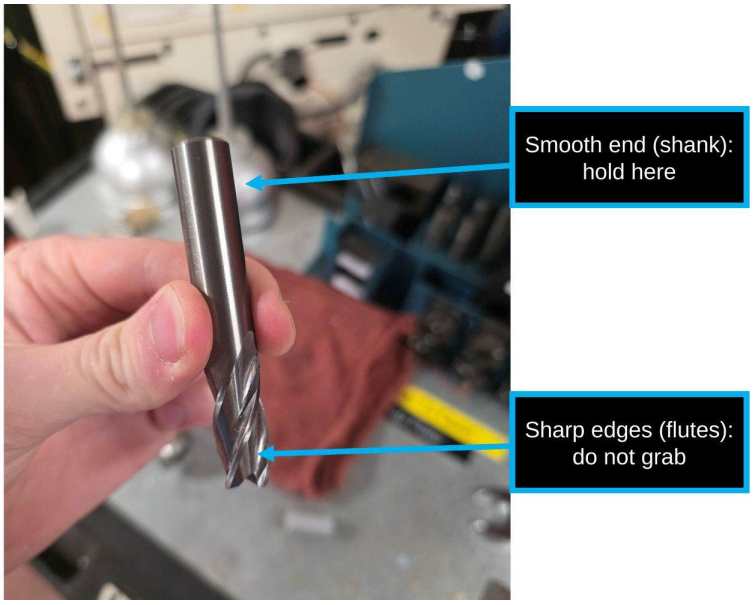


Figure 8. Hold the end mill by its smooth end (shank), never by the sharp cutting edges (flutes).

5. Hold the **collet** in your other hand with its narrow, slotted end facing the **end mill**. Push the **smooth end** of the **end mill** into that end of the **collet** until only the covered cutting end sticks out (Figure 10).

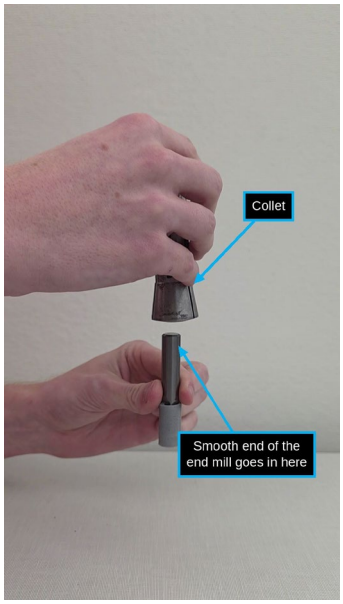


Figure 9. The smooth end of the end mill goes into the narrow, slotted end of the collet.

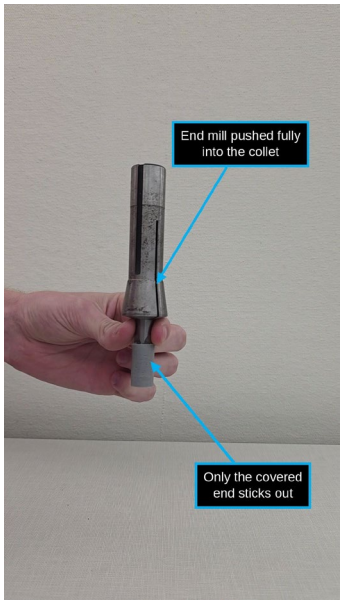


Figure 10. End mill pushed fully into the collet, with its protective cover still on.

6. Holding the **end mill and collet** together in one hand, push the **collet** all the way up into the **spindle**. Turn it slightly until it slides fully into place (Figure 11).

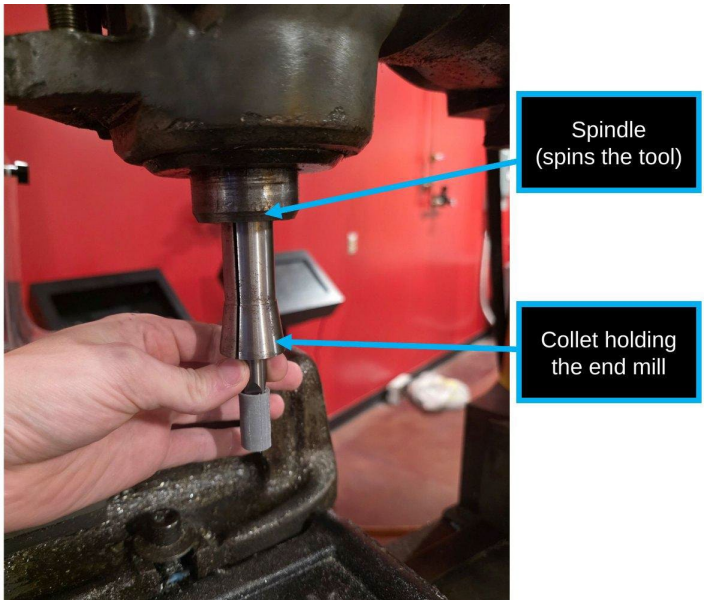


Figure 11. Pushing the collet and end mill up into the spindle.

7. Keep holding the **collet** from below with one hand. The **drawbar** is on the very top of the machine (Figure 12). With your other hand, turn the **drawbar** clockwise until it feels snug. If you cannot reach the **drawbar** while standing on the floor, ask your instructor for help. Do not stand on anything to reach it.

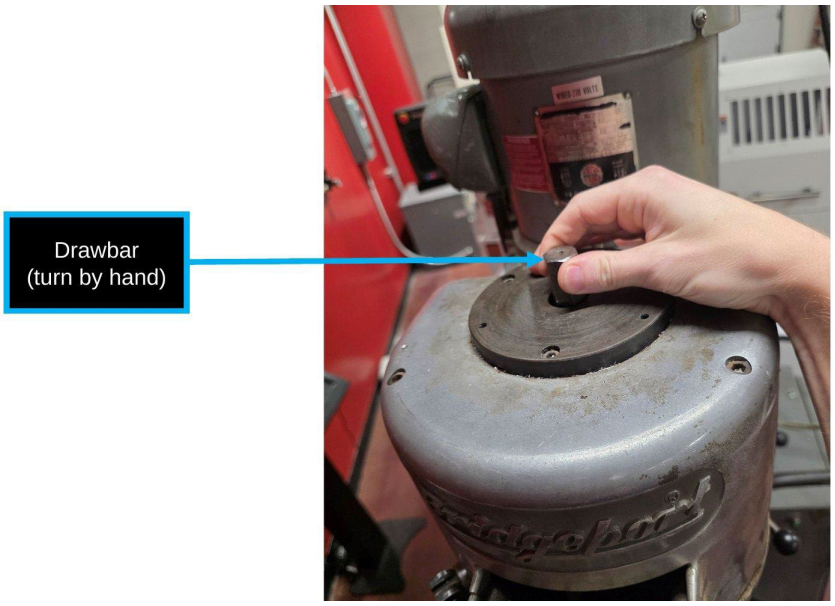


Figure 12. The drawbar is on the very top of the machine. Turn it by hand.

8. Find the **spindle brake lever** on the upper part of the machine (Figure 13). Hold it firmly with one hand so the **spindle** cannot turn. It takes more force than you expect. If you cannot reach the **spindle brake** while standing on the floor, ask your instructor for help.

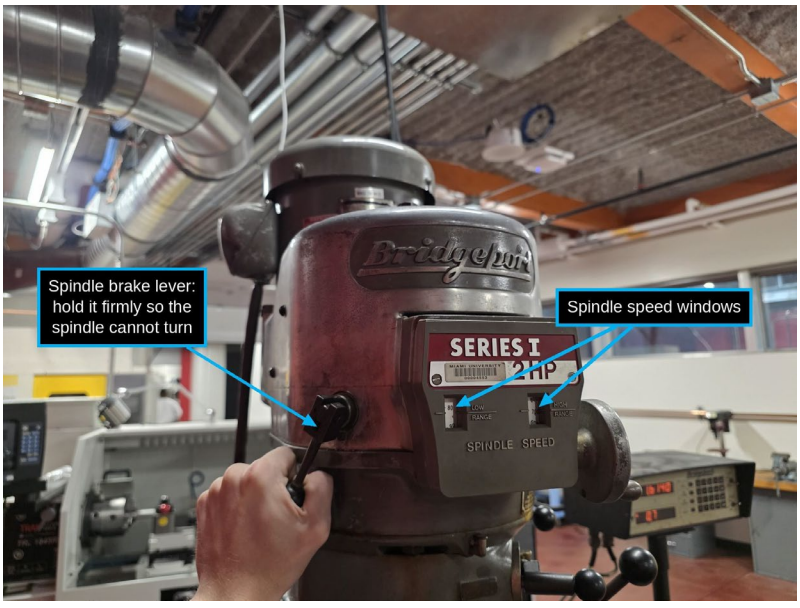


Figure 13. Spindle brake lever and spindle speed windows on the upper part of the machine.

9. While holding the **spindle brake**, use the **drawbar wrench** in your other hand to tighten the **drawbar** clockwise until it is snug (Figure 14). Then let go of the **Spindle brake.**

    - If you cannot hold the brake and reach the **drawbar** with the **wrench** at the same time, ask your instructor to help with one of them.

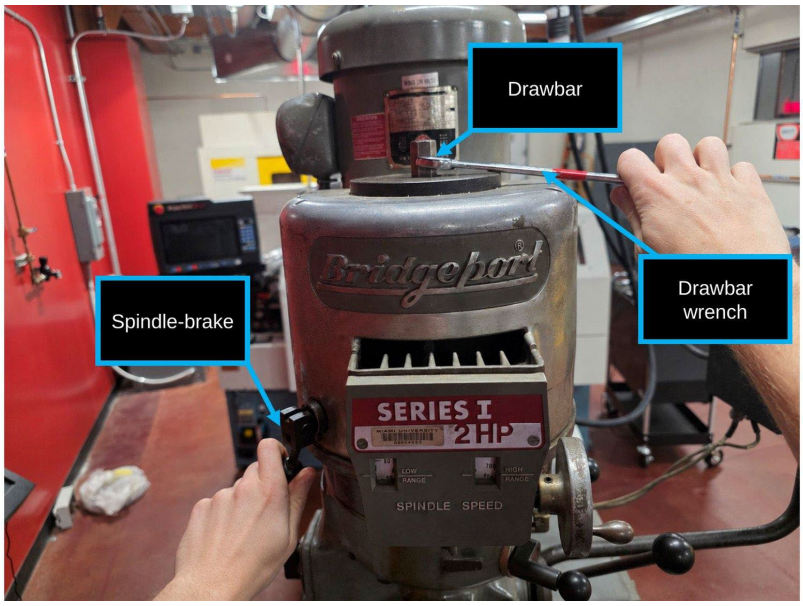


Figure 14. Holding the spindle brake while tightening the drawbar with the drawbar wrench.

10. Make sure the **wrench** is off the **drawbar** back on the **toolbox**.

11. Hold the **end mill** by its **protective cover** and pull down gently; the endmill should not move or drop.

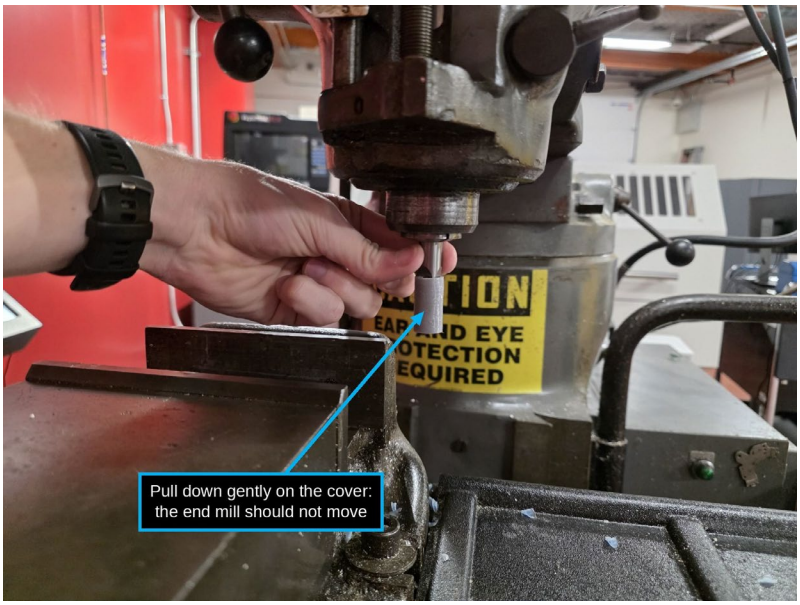


Figure 15. Checking that the end mill is tight.

Module 3: Loading the Wax Workpiece

AR System Guidance

- Maintain the main-power OFF condition from the preceding module and prompt confirmation that the **spindle** is stopped before **workpiece** handling. Display a reminder to keep the **protective cover** on the **end mill** during loading.
- Highlight the **vise handle** on the **toolbox** and the matching attachment point on the **vise**. Display Figure 16 for component identification and prompt the operator to attach the **handle** and open the **vise** wide enough for the **parallels** and **wax block**. Do not guess a **handle** rotation direction.
- Highlight the **chip brush** and the **vise interior**. Display Figure 17 and prompt the operator to **brush** out **chips**. Display the warning not to clear **chips** with hands.
- Highlight the two **parallels** and show their placement in the **vise**, one against each side, using Figure 18. Explain with the source wording that they support the **wax block** above the **vise** and keep it level.
- Highlight the **wax block** and the top surfaces of the **parallels**. Display Figures 19 and 20 and show the **block** placement on both **parallels**. Prompt the operator to turn the **vise handle** to clamp the **block** tight while keeping the **cutter** covered.
- Display Figure 21 and prompt the operator to perform the source wiggle check by hand. Ask them to confirm that the **block** is flat on both **parallels** and does not move; if movement is reported, prompt further **vise** tightening and repeat the confirmation before continuing.
- Highlight the **vise handle** and **toolbox**. Prompt the operator to remove the **handle** and return it to the **toolbox** so it cannot be hit during cutting. Require explicit confirmation that the **handle** is removed.
- Allow the next module only after the operator confirms correct seating on the **parallels**, secure clamping, the **protective cover** still on, and the **vise handle** removed. Do not treat a matching photograph as proof that the **block** is tight.

[REFERENCE IMAGES: Figures 16 to 21 show vise-handle identification, brushing, parallel placement, workpiece loading, and the clamp check. Figure 1 shows the return location for the vise handle.]

[NEEDS CONFIRMATION: The final loading instruction refers to Figure 16 as the finished setup, but Figure 16 shows the vise handle attached. The written instruction requires the handle to be removed. Do not use that handle position as a finished-setup overlay; obtain an updated reference image.]

Operator Procedure

1. Take the **vise handle** from the **toolbox** and put it on the **vise** (Figure 16). Turn it to open the **vise** wide enough for the **parallels** and the **wax block**.

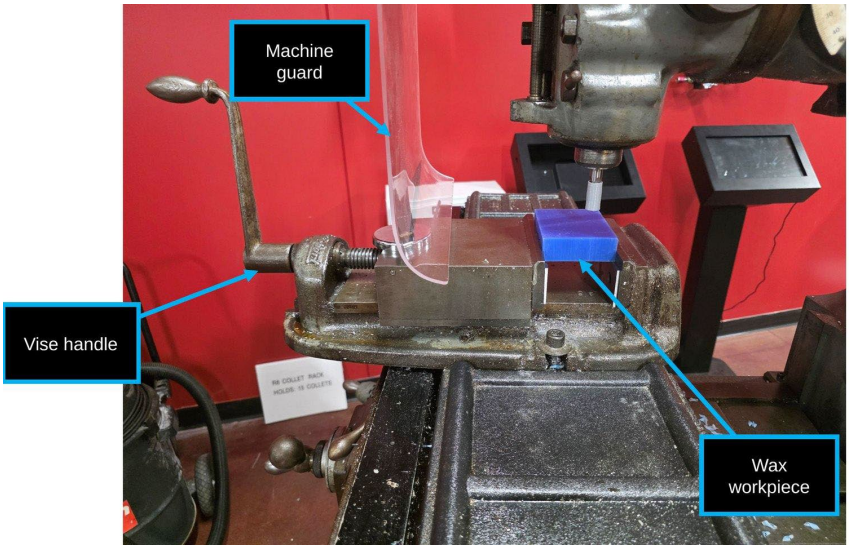


Figure 16. Vise handle on the vise, machine guard, and wax workpiece.

2. **Brush** any **chips** out of the **vise** with the **chip brush**. Never clear **chips** with your hands.

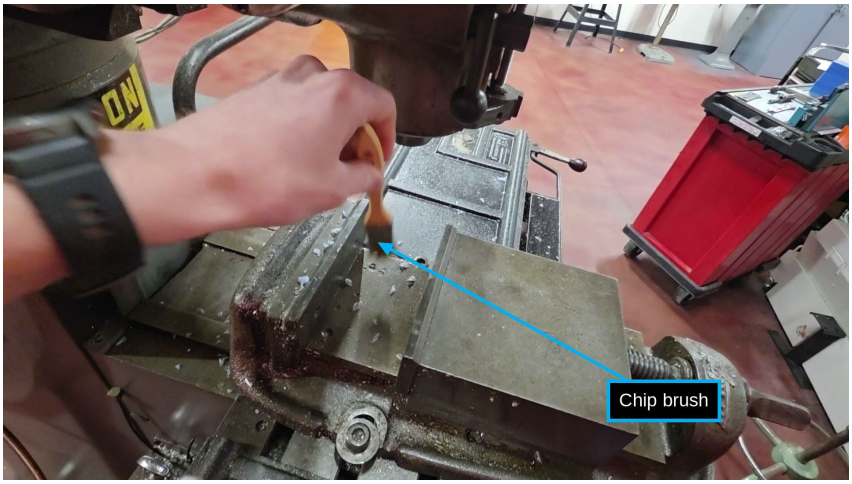


Figure 17. Brushing chips out of the vise with the chip brush.

3. Stand the two **parallels** in the bottom of the **vise**, one against each side. The **parallels** lift the **wax block** so its top sits above the **vise** and keep it level.



Figure 18. The two parallels standing in the vise, one against each side.

4. Set the **wax block** on top of the **parallels**, then turn the **vise handle** to clamp the **block** tight (Figures 19 and 20). Keep the **protective cover** on the **end mill** while you work near it.

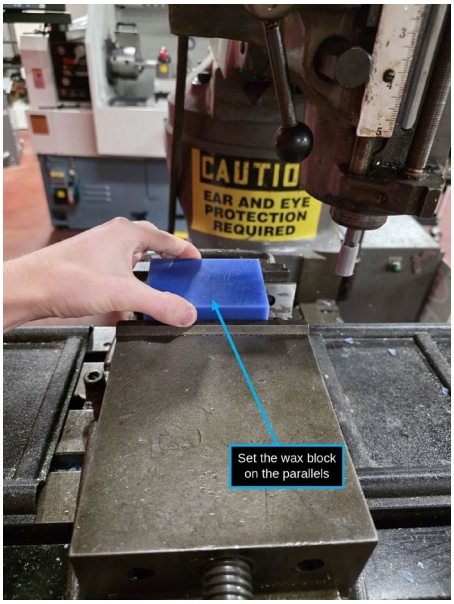


Figure 19. Setting the wax block on top of the parallels.

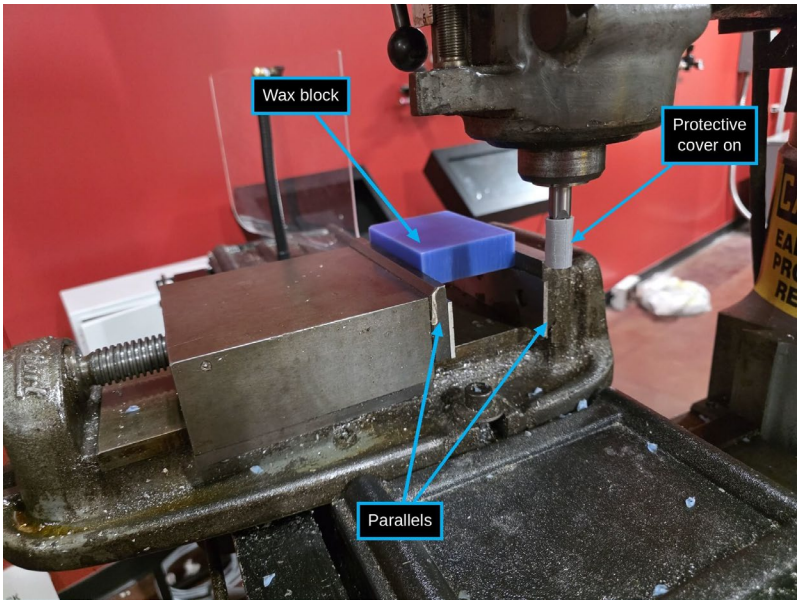


Figure 20. Wax workpiece clamped on the parallels. The protective cover stays on the end mill while loading.

5. Check the **block** with your hand: try to wiggle it (Figure 21). It should sit flat on both **parallels** and not move. If it moves, tighten the **vise** more.

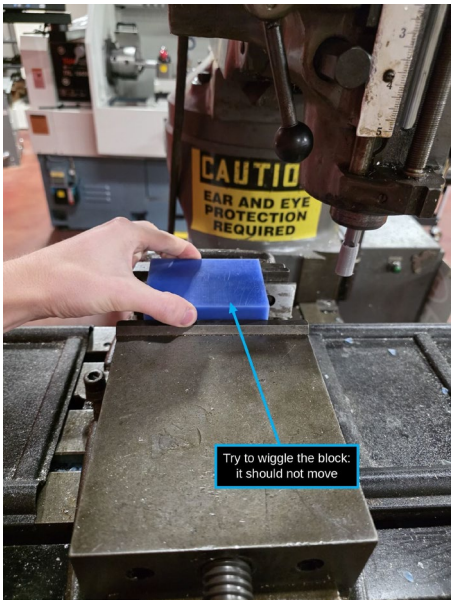


Figure 21. Trying to wiggle the wax block to make sure it is clamped tight.

6. Take the **vise handle** off the **vise** and put it back on the **toolbox** (Figure 1), so it cannot be hit while cutting.

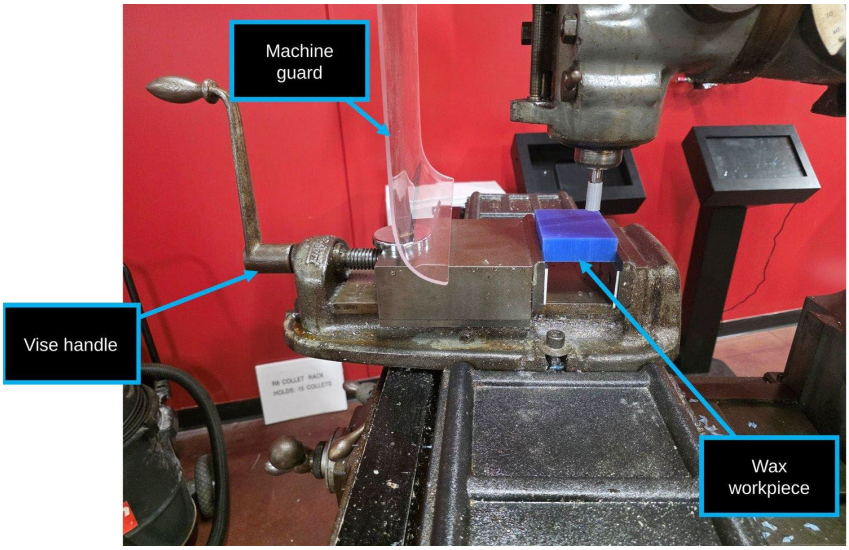


Figure 16. Vise handle on the vise, machine guard, and wax workpiece.

Module 4: Preparing the Cut and Positioning the Guard

AR System Guidance

- Display the finished example in Figure 22 and **cutting map** in Figure 26. Label the outer-edge cutting route and **central raised area**. State that the goal is **handwheel** practice and that the source does not require a perfect finished shape.
- Require confirmation that power remains OFF and the **spindle** is stopped before **cover** removal or touching off. Highlight the **protective cover** and display Figures 23 and 24. Prompt the operator to hold the **cover** only, twist it along the spiral edges, and pull it off while keeping fingers away from the **sharp edges**.
- Highlight the **toolbox** and prompt the operator to place the removed **cover** there. Do not specify clockwise or counterclockwise for the **cover** twist because the procedure does not name a direction.
- Highlight the **X-axis and Y-axis handwheels** and prompt the operator to move the **wax block** under the **end mill**. Label the **block** and **cutter tip**; do not add a numeric alignment offset or an unspecified number of turns.
- Highlight the **Z-axis crank** and display a clockwise raising arrow consistent with Figure 5. Display Figure 25 and prompt slow raising until the **cutter tip** just touches the top of the **wax block**. Prompt the operator to stop at first contact and confirm this as Z-axis zero.
- After touching off, keep a reminder displayed: do not turn the **Z-axis crank** again during preparation and cutting. End this reminder only when the shutdown step specifically instructs lowering the **table**.
- Display Figure 26 with front/back orientation and S, A, B, C, and D labeled. Prompt the operator to use the **X and Y handwheels** to position the **cutter** off the side of the **block** next to A, at the start location S. Use Figure 27 as a position reference and require confirmation that the **cutter** is clear of the **block**.
- Highlight the **machine guard** and its stem using Figure 28. Prompt the operator to move the **guard** in front of the **cutting area** while the **spindle** is stopped and confirm that it will not obstruct **table** travel.
- Display Figure 29 for **guard** and uncovered-**cutter** identification. Require confirmation that the **cutter cover** is removed, the **guard** is in position, the **workpiece** remains secure, and the **cutter** is clear of the **block** before allowing startup.

[REFERENCE IMAGES: Figures 22 to 29 show the finished example, cover removal, touching off, cutting route, off-block start position, and guard placement.]

[NEEDS CONFIRMATION: Figure 29 shows a vise handle attached and its caption calls the setup ready to cut, while the loading procedure requires the handle to be removed. Do not use the attached handle as an approved ready-to-cut condition.]

Operator Procedure

1. The goal is to cut around the outer edges of the **wax block**, leaving a **raised area** in the middle (Figure 26). Your cut does not have to be perfect; the point is to practice moving the **table** with each **handwheel**. Figure 22 shows an example of a finished **block**.

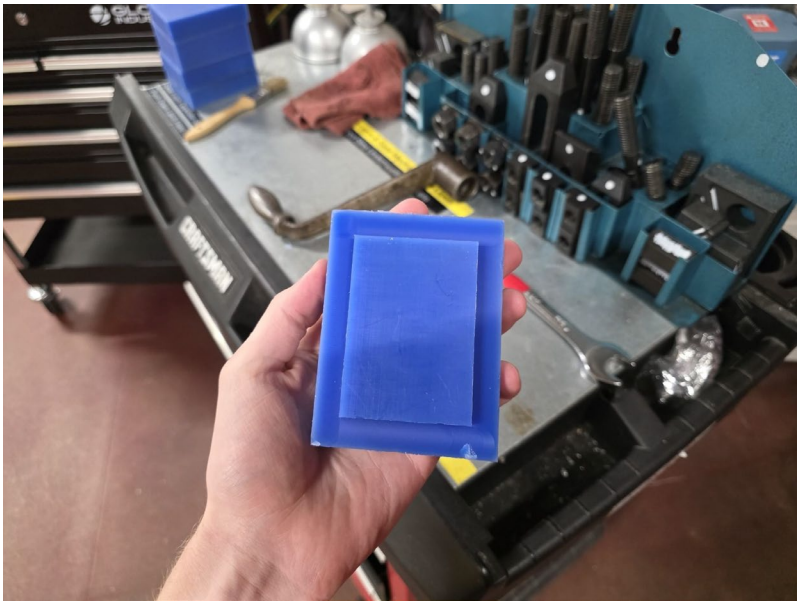


Figure 22. Finished example: the outer edges are cut away, leaving a raised area in the middle.

2. Remove the **protective cover** from the **end mill**. Hold only the **cover**, twist it so it follows the spiral edges, and pull it off (Figures 23 and 24). Keep your fingers away from the **sharp edges**. Put the **cover** on the **toolbox**.

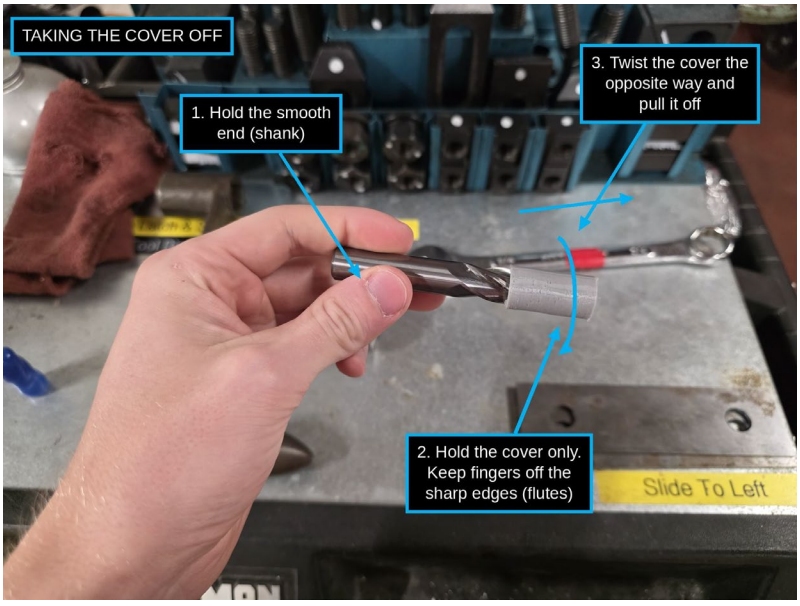


Figure 23. Taking the protective cover off: twist it and pull.

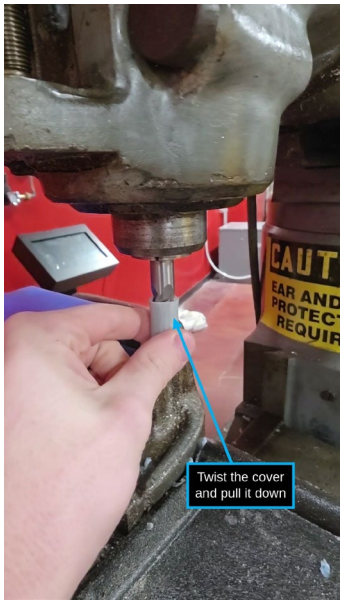


Figure 24. Removing the protective cover at the machine.

3. Use the **X-axis and Y-axis handwheels** to move the **wax block** under the **end mill**. Then slowly raise the **table** with the **Z-axis crank** until the tip of the **end mill** just touches the top of the **wax block** (Figure 25). Stop there. This is called touching off: it sets the zero height (Z-axis zero), so the cut depth is measured from the top of the **block**. Do not turn the **Z-axis crank** again after this.

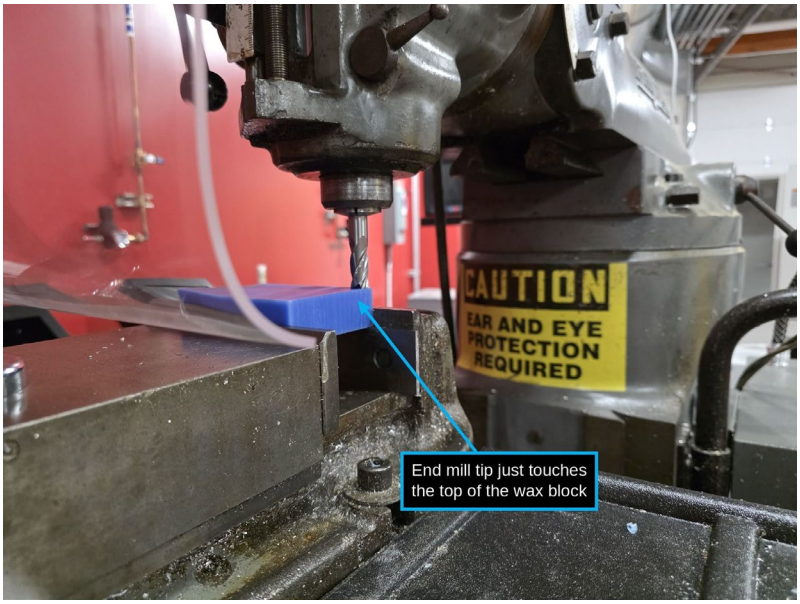
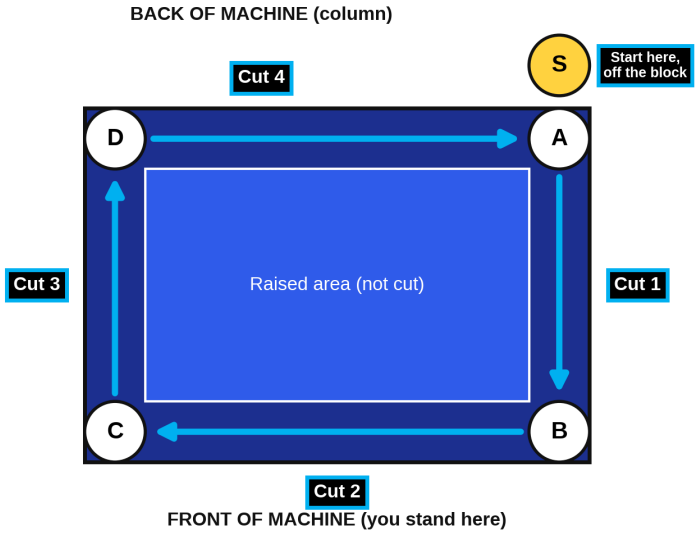


Figure 25. The tip of the end mill just touching the top of the wax block.

4. Use the **handwheels** to move the **end mill** off the side of the **block**, next to point A on the **cutting map** (Figures 26 and 27). The **end mill** should be clear of the **block**.



Top view. Each circle is the size of the end mill. Start at S, off the block, then cut through A to B, C, D and back to A.

Figure 26. Cutting map (top view of the wax block). Start at S, off the block. Cut through A to B, then B to C, C to D, and D back to A.

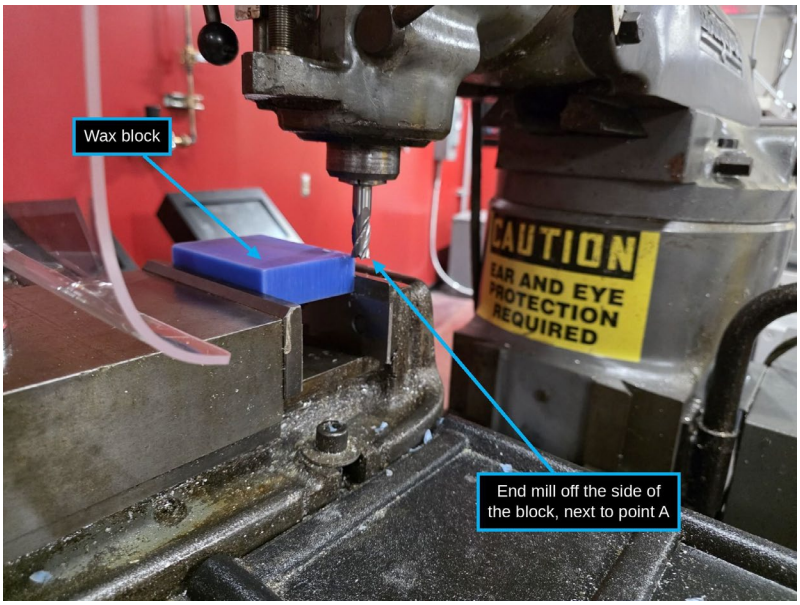


Figure 27. End mill off the side of the wax block, next to point A.

5. Move the **machine guard** into place in front of the **cutting area** (Figures 28 and 29). The **guard** protects you from flying **chips** and from a **tool** or part that comes loose. Make sure it does not **block** the **table** from moving.

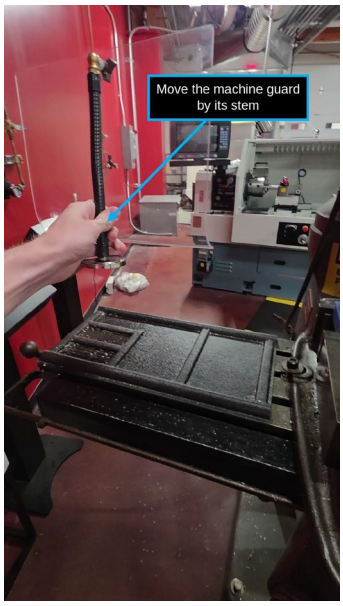


Figure 28. Moving the machine guard into place.

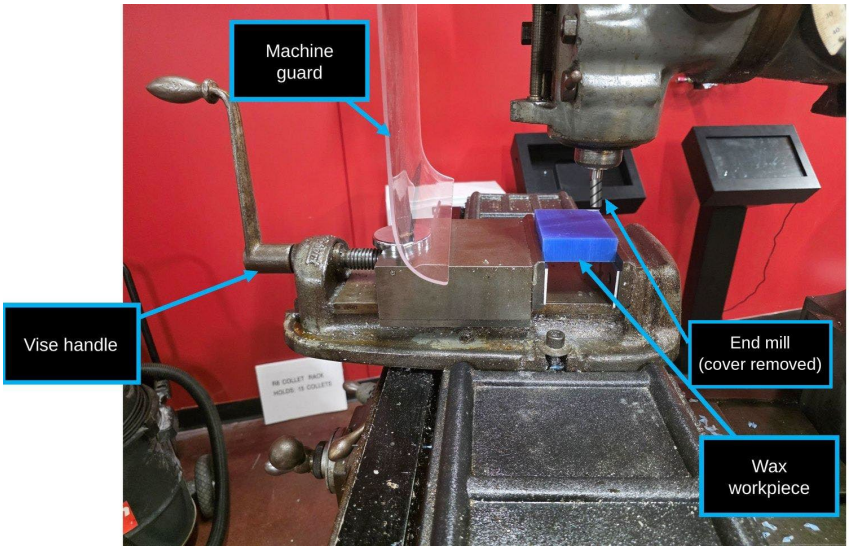


Figure 29. Finished setup, ready to cut: machine guard in place and protective cover removed.

Module 5: Machine Startup

AR System Guidance

- Before presenting a power-on prompt, require operator confirmation that the **workpiece** is clamped, the **guard** is in place, the **end mill** is clear of the **block**, everyone is standing clear, and the **vise handle** and **drawbar wrench** are returned to the **toolbox**.
- Highlight the **spindle speed windows** using Figure 13. Display the expected source settings exactly: HIGH range and about 700 RPM. Prompt the operator to check them. If different or unreadable, pause the workflow and direct the operator to talk to the instructor; do not provide novice speed-adjustment instructions.
- Highlight and label the **spindle direction switch**, **Spindle ON button**, and red **E-STOP** on the front **control panel** using Figure 30. Prompt the operator to identify **E-STOP** before starting the **spindle**.
- Identify the **bottom-row table-feed controls** and display a Do not use reminder because the source procedure uses manual **table** movement. Do not highlight those buttons as actions to perform.
- Highlight the **main power switch** and display Figure 31 showing ON. Prompt the operator to turn it to ON only after the setup checks have been confirmed.
- Highlight the **spindle direction switch** and show the CW setting using Figure 30. Prompt the operator to set CW and display a reminder not to press **Spindle ON** yet. For rotation verification, state the source viewpoint: clockwise viewed from above the **spindle**.
- Allow the machining module after the startup checks and CW setting are confirmed. Keep **Spindle ON** as a later action, after the operator has read the full machining and stopping sequence.

[REFERENCE IMAGES: Figure 13 identifies the speed windows; Figure 30 identifies the panel controls; Figure 31 shows main-power ON.]

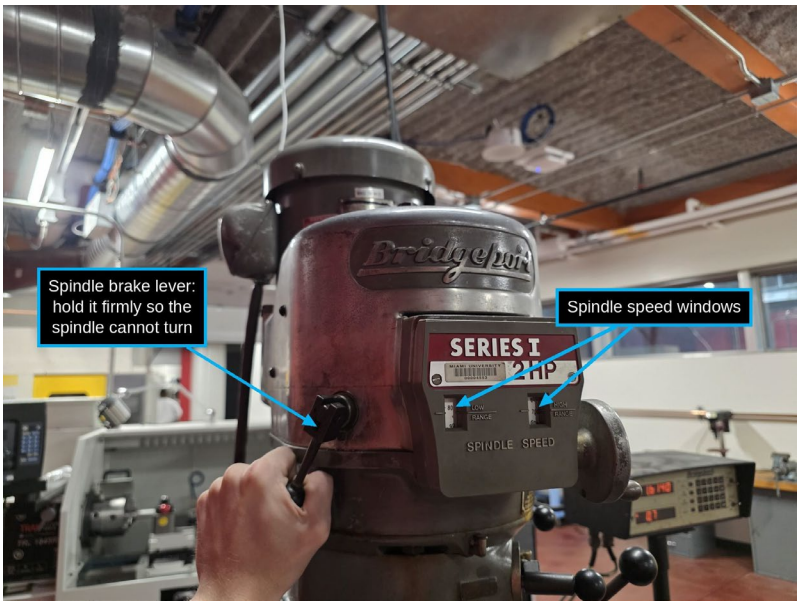


Figure 13. Spindle brake lever and spindle speed windows on the upper part of the machine.

Operator Procedure

1. Check that the **workpiece** is clamped, the **guard** is in place, the **end mill** is not touching the **block**, and everyone is standing clear.

2. Look at the **spindle speed windows** (Figure 13). The machine should be in HIGH range and the speed should read about 700 RPM. If it is different, talk to your instructor. Do not change the speed yourself.

3. Find the **control panel** on the front of the machine and identify the **spindle direction switch**, the **Spindle ON button**, and the red **E-STOP** (emergency stop) button (Figure 30). Do not use the buttons on the bottom row; they move the **table** with a motor.

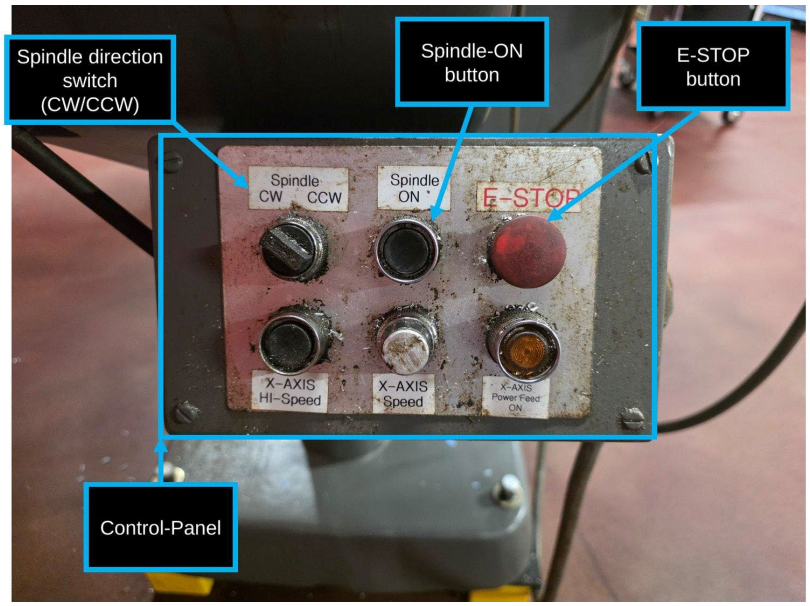


Figure 30. Control panel on the front of the machine: spindle direction switch, Spindle ON button, and E-STOP button.

4. Turn the **main power switch** to ON (Figure 31).



Figure 31. Main power switch turned to ON.

5. Set the **spindle direction switch** to CW (clockwise) (Figure 30). Do not press **Spindle ON** yet.

Module 6: Machining the Part and Powering Off

AR System Guidance

- Display the complete source machining section, direction **table**, cutting-step **table**, and Figure 26 before presenting **Spindle ON**. Require acknowledgment that the operator has read the entire cutting and stopping sequence.
- Highlight only the **X-axis handwheel** at the right end for all X-axis cutting moves. Display the source warning that the left-end **handwheel** turns the opposite way. Show the relevant row of the direction **table** before each move.
- Highlight the **quill feed handle**, **quill stop**, and **depth scale** using Figures 32 and 33. Display the source depth of 0.25 in. and the Figure 32 annotation that the stop is set by the instructor. Prompt confirmation of the instructor-set stop; do not add student stop-adjustment instructions or infer the depth from an uncalibrated visual overlay.
- For cutting step 1, display S next to A on Figure 26 and Figure 27. Highlight the **X and Y controls** as needed and prompt confirmation that the **cutter** is off the side of the **block** at S before starting.
- For cutting step 2, highlight **Spindle ON** and prompt the operator to press it. Display a rotating-**cutter** warning and the **E-STOP** location before the action. Prompt the operator to observe the **spindle** direction and listen or feel for abnormal behavior without approaching rotating parts. If rotation is wrong or noise or shaking is abnormal, display Press **E-STOP** right away and pause progression.
- For cutting step 3, highlight the **quill feed handle** and show a downward movement arrow. Display Figure 34 and prompt the operator to pull the **handle** until it touches the **quill stop**, lowering the **end mill** 0.25 in. Then prompt the operator to let go, as the source specifies, and not touch the **handle** again until the cut is finished.
- For Cut 1, highlight route S through A to B on the source map and highlight the **Y-axis handwheel**. Display a clockwise rotation arrow and the separate **table**-motion label **Table** moves in, away from you. Keep the source distance field ___ turns; do not invent a value or a live remaining-turn count.
- For Cut 2, highlight B to C and the **right-end X-axis handwheel**. Display a counterclockwise arrow and **Table** moves right. Preserve ___ turns as the source distance field.
- For Cut 3, highlight C to D and the **Y-axis handwheel**. Display a counterclockwise arrow and **Table** moves toward you. Preserve ___ turns as the source distance field.
- For Cut 4, highlight D back to A and the **right-end X-axis handwheel**. Display a clockwise arrow and **Table** moves left. Display the source distance instruction Same as step 5, meaning the turn count used for Cut 2; do not confuse it with module numbering.
- Present the cutting rows one at a time and require operator confirmation of each completed segment before showing the next. Keep the full **cutting map** available. Explain through labels that the map shows the **cutter** path relative to the **block** while the **handwheel** arrows show actual **table** movement; do not invert the source control directions.
- Keep a safety zone visible but very transparent throughout cutting
- After Cut 4, highlight **E-STOP** and prompt the operator to press it. Display Wait until the **end mill** has completely stopped spinning and require operator confirmation of complete stop before presenting any **quill**-retraction, **guard**-opening, or cleaning action.
- Highlight the **quill feed handle** and show an upward arrow for moving it all the way back up. Then highlight the **Z-axis crank** and show a counterclockwise arrow for lowering the **table** several turns until the **end mill** is well above the **wax block**. This is the source-authorized end of the earlier do-not-turn-Z reminder.
- Highlight the **X and Y handwheels** and prompt the operator to move the **end mill** away from the **block** after lowering the **table**. Do not invent a clearance dimension or turn count for this repositioning.

- Highlight the **main power switch** and show Figure 6. Prompt the operator to turn it OFF and confirm OFF before the cleaning module. Do not treat main-power OFF as proof of a secured energy-isolation procedure.

[REFERENCE IMAGES: Figures 3 to 5 supply handwheel directions; Figure 26 supplies the cutting map; Figures 27, 30, and 32 to 34 supply the starting position, stop control, and quill operation.]

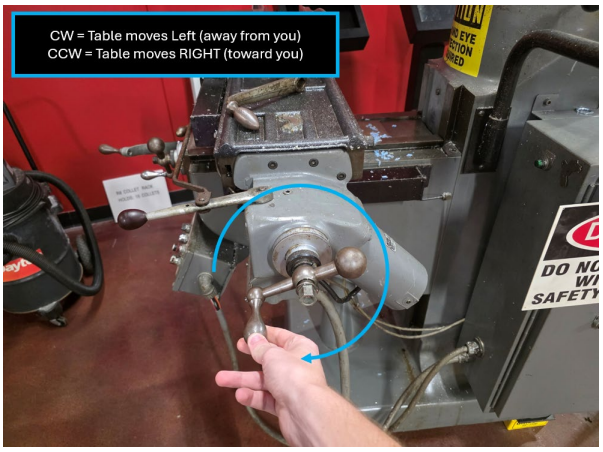


Figure 3. X-axis handwheel at the right end of the table: turning it clockwise moves the table to the left; counterclockwise moves it to the right.

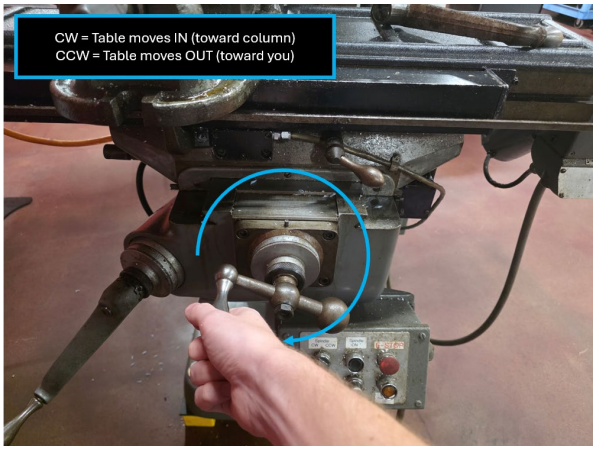


Figure 4. Y-axis handwheel: turning it clockwise moves the table in, away from you; counterclockwise moves it out, toward you.

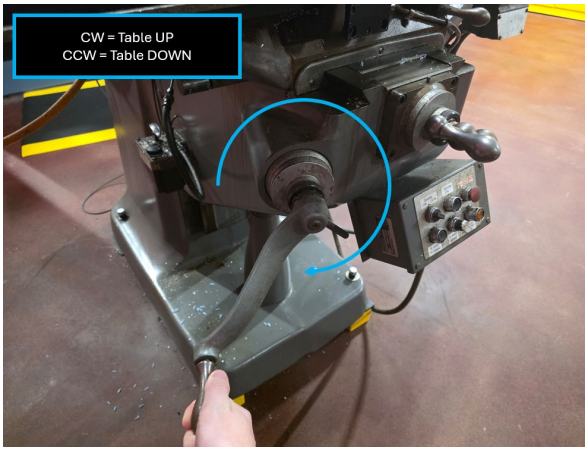
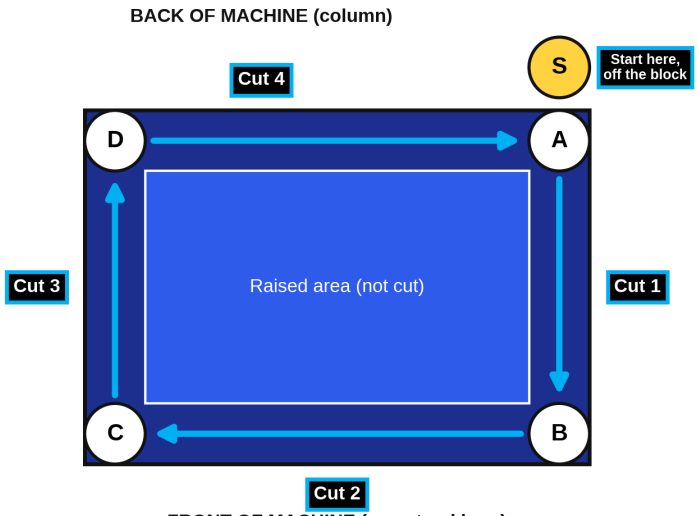


Figure 5. Z-axis crank: turning it clockwise raises the table; counterclockwise lowers it.



Top view. Each circle is the size of the end mill. Start at S, off the block, then cut through A to B, C, D and back to A.

Figure 26. Cutting map (top view of the wax block). Start at S, off the block. Cut through A to B, then B to C, C to D, and D back to A.

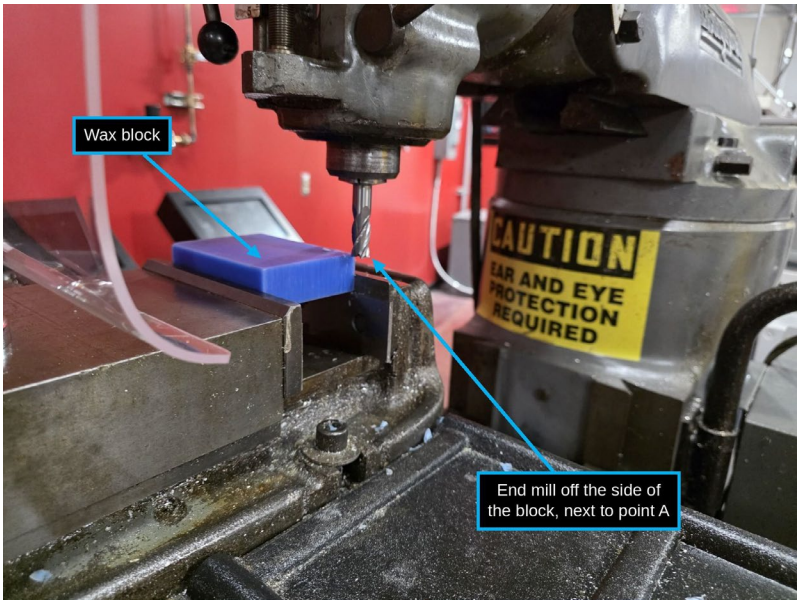


Figure 27. End mill off the side of the wax block, next to point A.

[NEEDS CONFIRMATION: Cutting steps 4, 5, and 6 contain blank ___ turns fields; step 7 uses Same as step 5. The instructor must provide approved turn counts or another explicit endpoint method before these rows can become executable AR cutting prompts. Preserve the original blanks in the Operator Procedure and block the affected cutting prompts until resolved.]

[NEEDS CONFIRMATION: Confirm with the instructor that the installed quill remains at the specified cutting depth after the operator releases the feed handle, as the SOP directs. No separate quill-lock action is specified; do not add one silently. Pause the affected cutting action until this behavior is confirmed.]

Operator Procedure

1. Read this whole section before you press **Spindle ON**, so you know every step of the cut and how to stop the machine when you are done.

2. Use the **X-axis handwheel at the right end of the table** (Figure 3) for all left and right moves. The **handwheel** at the left end turns the opposite way. Check the direction **table** below before each move.

Handwheel directions

Control	Turn clockwise (CW)	Turn counterclockwise (CCW)	See
X-axis handwheel (right end of table)	**Table** moves left	**Table** moves right	Figure 3
Y-axis handwheel (front, below **table**)	**Table** moves in, away from you	**Table** moves out, toward you	Figure 4
Z-axis crank (front-left, below **table**)	**Table** moves up	**Table** moves down	Figure 5

3. Find the **quill feed handle** on the right side of the upper part of the machine (Figure 32). Pulling it down lowers the spinning **end mill**. The **quill stop** is set so the **handle** stops when the **end mill** is 0.25 in. down (Figures 33 and 34).

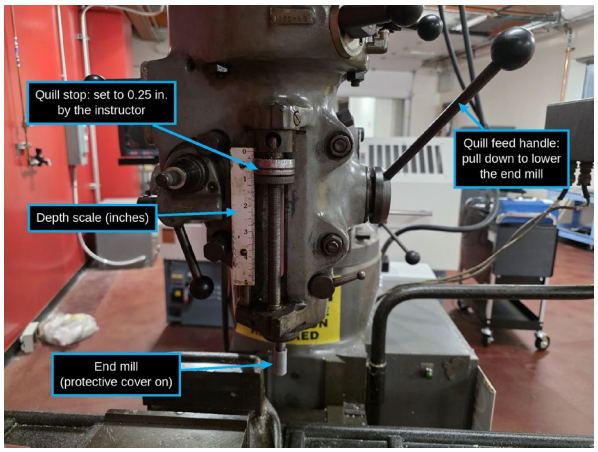


Figure 32. Quill feed handle, quill stop, and depth scale.

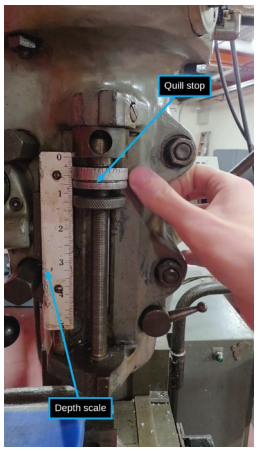


Figure 33. The quill stop and the depth scale.

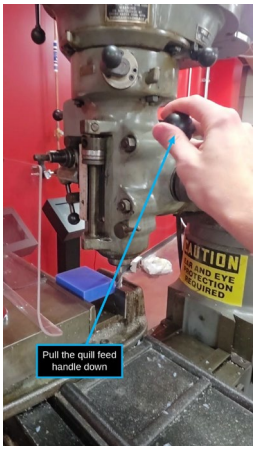


Figure 34. Pulling the quill feed handle down.

4. Follow the cutting steps below, one row at a time, using the **cutting map** (Figure 26).

Cutting steps

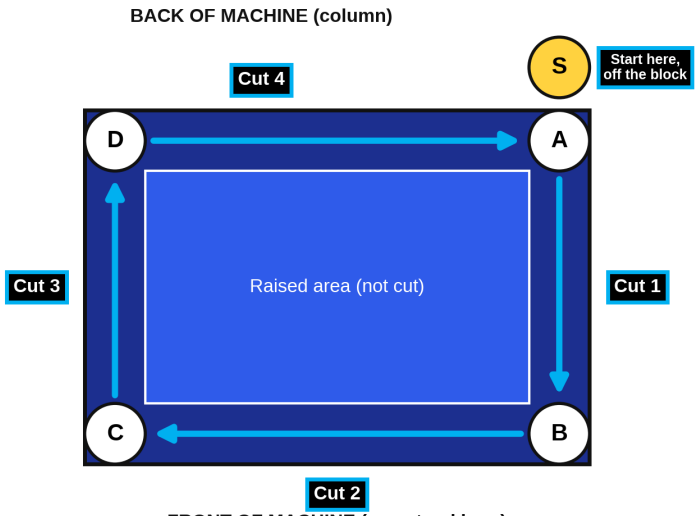


Figure 26. Cutting map (top view of the wax block). Start at S, off the block. Cut through A to B, then B to C, C to D, and D back to A.

Step	What to do	Control and direction	How far
5	Check that the **end mill** is at S, off the side of the **block** next to point A.	**X and Y handwheels**	As needed
5	Press the **Spindle ON button**. If the **spindle** turns the wrong way or sounds or shakes abnormally, press **E-STOP** right away.	**Spindle ON button**	

Step	What to do	Control and direction	How far
6	Pull the **quill feed handle** down until it touches the **quill stop**. This lowers the **end mill** 0.25 in. Then let go; you do not need to touch the **handle** again until the cut is finished.	**Quill feed handle**, down	Until it touches the **quill stop** (0.25 in.)
7	Cut 1: move the **end mill** into the **block** and through A to B.	**Y handwheel**, CW (**table** moves in, away from you)	___ turns
7	Cut 2: from B to C.	**X handwheel (right end)**, CCW (**table** moves right)	___ turns
7	Cut 3: from C to D.	**Y handwheel**, CCW (**table** moves toward you)	___ turns
7	Cut 4: from D back to A.	**X handwheel (right end)**, CW (**table** moves left)	Same as step 5
8	Press **E-STOP** and wait for the **end mill** to stop spinning.	**E-STOP button**	

19 Move the **quill feed handle** all the way back up (Figure 32).

10. Lower the **table**: turn the **Z-axis crank** counterclockwise several turns (Figure 5) until the **end mill** is well above the **wax block**. This leaves room for the next person to load a new **block**. Then use the **X-axis and Y-axis handwheels** to move the **end mill** away from the **block**.

11. Turn the **main power switch** to OFF (Figure 6).



Figure 6. Main power switch on the gray electrical box, turned to OFF.

Module 7: Cleaning the Work Area and Unloading

AR System Guidance

- Require confirmation that the **spindle** and all machine movements have stopped and main power is OFF before presenting cleaning actions. If motion is reported, keep the workflow paused and display a wait-for-stop warning.
- Highlight the **machine guard** and prompt the operator to move it out of the way. Display Figure 35 for **guard** location and a reminder not to touch the **end mill**.
- Highlight the **shop vacuum** storage location using Figure 36. Prompt the operator to retrieve the **vacuum** and plug it into the **nearest outlet** as stated in the SOP. Do not assign an **outlet** that the source does not identify.
- Highlight the **vacuum ON/OFF switch** using Figure 37 and prompt the operator to turn it on. Highlight the **chips** on the **workpiece**, **vise**, and **table** using Figures 38 and 39 and prompt vacuuming of those surfaces and any **chips** on the floor.
- Keep a sharp-**tool** reminder visible: keep hands and the **vacuum hose** away from the **end mill**. Display the source prohibitions against removing **chips** by hand or with compressed air.
- Prompt the operator to turn the **vacuum** OFF, unplug it, and return it to its storage location. Require confirmation that it is returned before unloading.
- Highlight the **vise handle** and its attachment point. Prompt the operator to refit the **handle** and loosen the **vise** without specifying an unsupported rotation direction.
- Highlight the **wax block** and **parallels**. Prompt removal without touching the **end mill**, then direct attention to the **toolbox** for returning the **parallels**. Prompt the operator to remove the **vise handle** again and return it to the **toolbox**.
- Allow the **tool**-removal module only after the operator confirms the work area is cleaned, the **vacuum** is off and returned, the **workpiece** and **parallels** are removed, and the **vise handle** is returned.

[REFERENCE IMAGES: Figures 35 to 39 show the opened guard, vacuum storage and controls, and chip-cleaning locations. Figure 44 identifies toolbox return locations.]

[NEEDS CONFIRMATION: Figure 35 shows the end mill with its protective cover on, but the written sequence replaces the cover in the following module. Use Figure 35 only to locate the guard; do not insert an earlier cover-fitting action without confirming the intended sequence.]

Operator Procedure

1. After the machine has stopped and the power is off, move the **machine guard** out of the way (Figure 35). Do not touch the **end mill**.

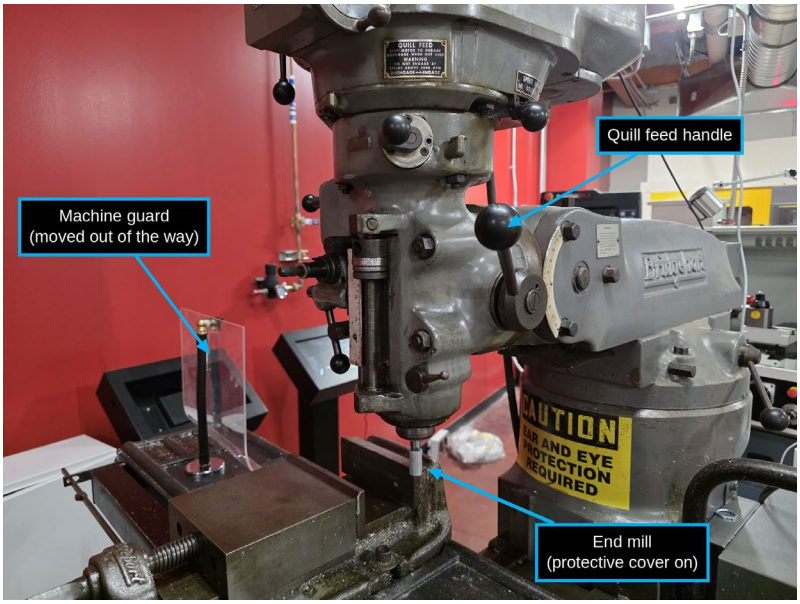


Figure 35. Machine guard moved out of the way, and the quill feed handle.

2. Get the **shop vacuum** from where it is stored next to the mill and plug it into the **nearest outlet** (Figure 36).



Figure 36. Where the shop vacuum is stored next to the mill.

3. Turn the **vacuum** on (Figure 37) and **vacuum** the **wax chips** off the **workpiece**, **vise**, and **table** (Figures 38 and 39). Also **vacuum** any **chips** on the floor. Keep your hands and the **vacuum hose** away from the **end mill**. Do not remove **chips** by hand or with compressed air.



Figure 37. Shop vacuum ON/OFF switch.

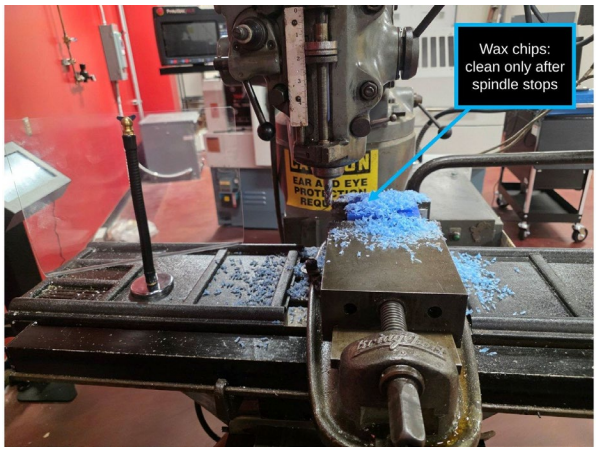


Figure 38. Wax chips on the vise and table after cutting.

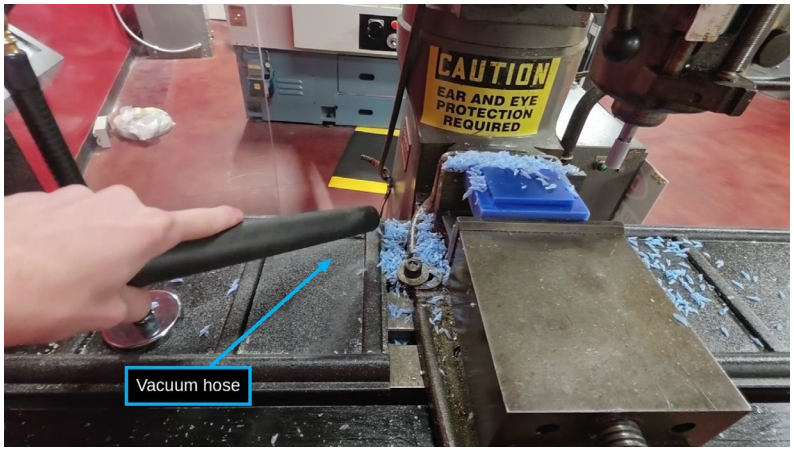


Figure 39. Vacuuming the wax chips off the vise and table.

4. Turn the **vacuum** off, unplug it, and put it back where you found it.

5. Put the **vise handle** back on the **vise** and loosen the vise.

6. Take out the **wax block** and the **parallels** without touching the **end mill**, and put the **parallels** back on the **toolbox**.

7. Take the **vise handle** off the **vise** and put it back on the **toolbox**.

Module 8: Removing the End Mill and Collet and Returning Tools

AR System Guidance

- Highlight the **main power switch** and require confirmation that it remains OFF and the **spindle** is completely stopped before handling the **cutter** or **drawbar**.
- Highlight the **protective cover** on the **toolbox** and display Figures 40 and 41. Prompt the operator to use two hands, steady the **cover**, align its internal grooves with the spiral edges, and twist and push it on. Do not invent a clockwise or counterclockwise twist direction.
- Highlight the **shop rag** and the **vise** beneath the **end mill**. Display Figure 42 and prompt the operator to place the **rag** under the **tool** before loosening the **drawbar**, so the **tool** is protected if it drops.
- Highlight the **spindle brake** and **drawbar wrench** using Figures 13 and 14. Prompt the operator to hold the brake firmly and show a counterclockwise arrow for loosening the **drawbar** with the **wrench**.
- Display the floor-standing reach reminder before the brake and **drawbar** actions. If either cannot be reached while standing on the floor, pause progression and prompt instructor assistance. Display Do not stand on anything to reach them.
- Once the **drawbar** is loose, highlight the **toolbox** and prompt the operator to return the **wrench**. Highlight the **covered end mill** as the support location and the **drawbar** in Figure 12; prompt support of the **tool** with one hand while unscrewing the **drawbar** the rest of the way by hand.
- Display that the **collet and end mill** should release into the supporting hand or onto the **rag** once the **drawbar** is free. Prompt the operator to confirm release before presenting the separation step.
- If the operator reports that the **collet** is still stuck, display the source action and Figure 43: lift the **drawbar** a couple of inches and let it drop to knock the **collet** loose. Do not add a hammer, a precise height, repeated-force instructions, or another recovery method. If it does not release, display Stop and ask your instructor for help and pause progression.
- Highlight the **end mill shank** and **collet**. Prompt the operator to remove the **end mill** from the **collet** by its **smooth end** and keep the **protective cover** on. Display the sharp-edge warning.
- Display Figure 44 and highlight the **toolbox** return positions for the **collet**, **covered end mill**, **drawbar wrench**, **vise handle**, **parallels**, **shop rag**, and **chip brush**. Prompt the operator to return every item to its original place and compare the **toolbox** with the reference photograph before leaving.
- Complete the AR procedure only after the operator confirms main power remains OFF, the **end mill and collet** are removed, the **cover** remains on the **end mill**, and all listed tools are returned. If a **tool** remains stuck or another unresolved condition is reported, keep the procedure paused for instructor assistance.

[REFERENCE IMAGES: Figures 6, 12 to 14, and 40 to 44 show OFF, drawbar and brake locations, cover replacement, rag placement, the source stuck-collet action, and final tool layout.]

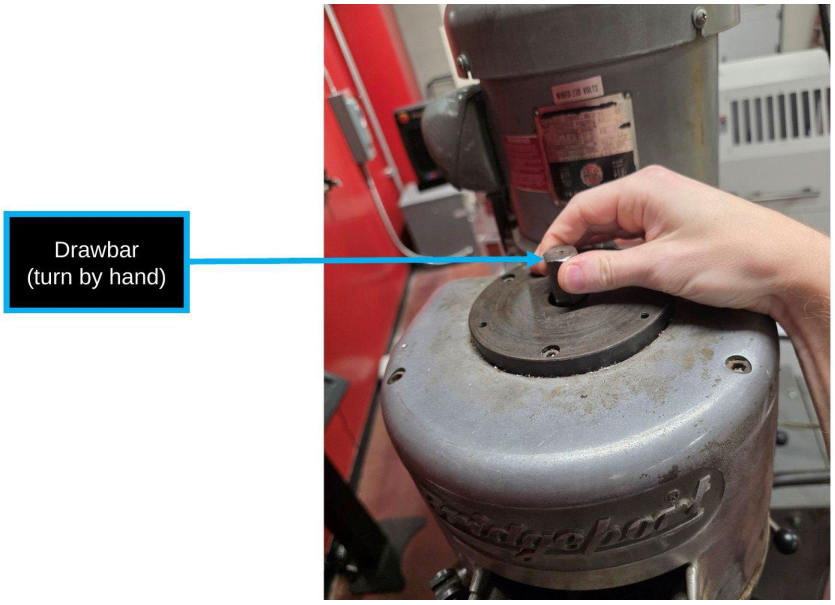


Figure 12. The drawbar is on the very top of the machine. Turn it by hand.

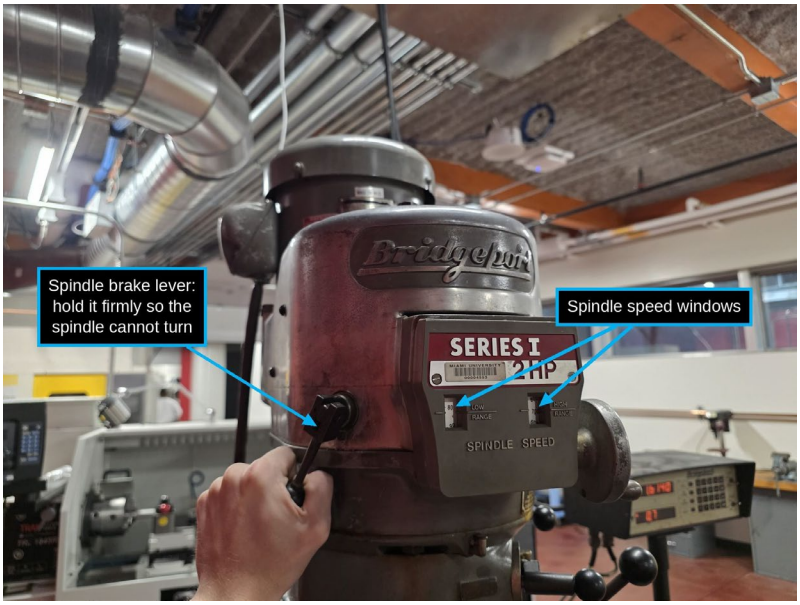


Figure 13. Spindle brake lever and spindle speed windows on the upper part of the machine.

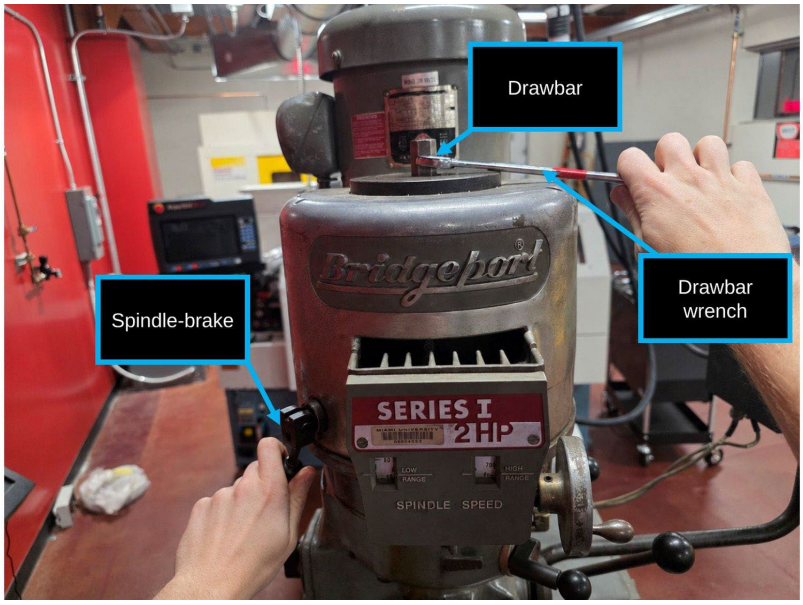


Figure 14. Holding the spindle brake while tightening the drawbar with the drawbar wrench.

Operator Procedure

1. Put the **protective cover** back on the **end mill** using two hands: steady the **cover** with one hand, line up the grooves inside the **cover** with the spiral edges, then twist and push it on (Figures 40 and 41).

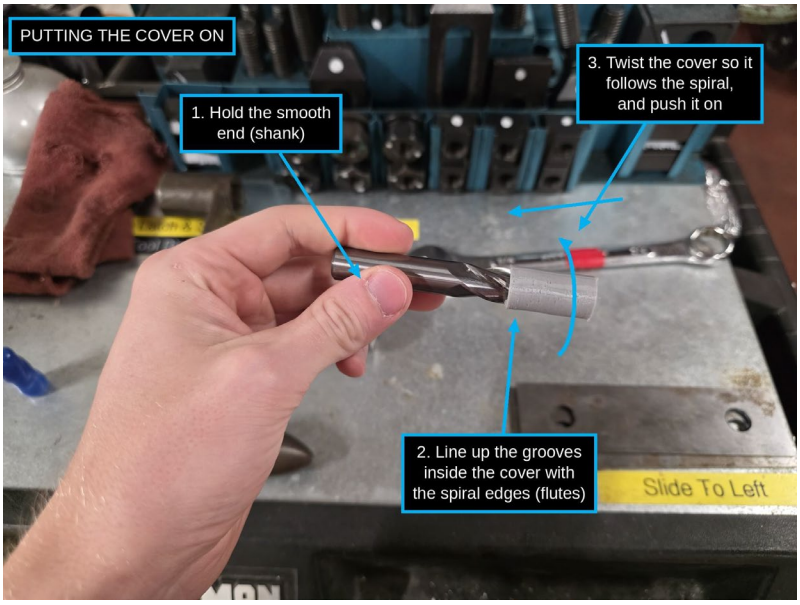


Figure 40. Putting the protective cover back on: line up the grooves, twist, and push.

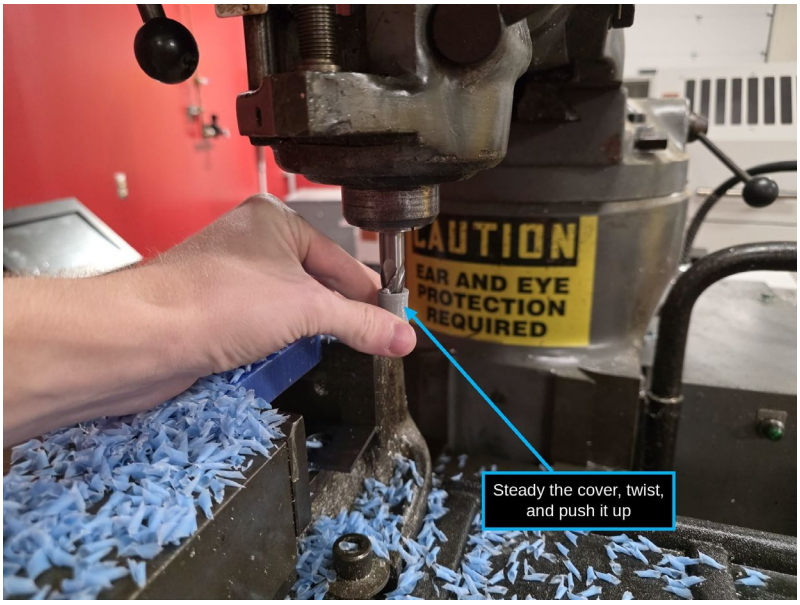


Figure 41. Putting the protective cover back on at the machine.

2. Place a **shop rag** on the **vise** under the **end mill** so the **tool** is not damaged if it drops (Figure 42).

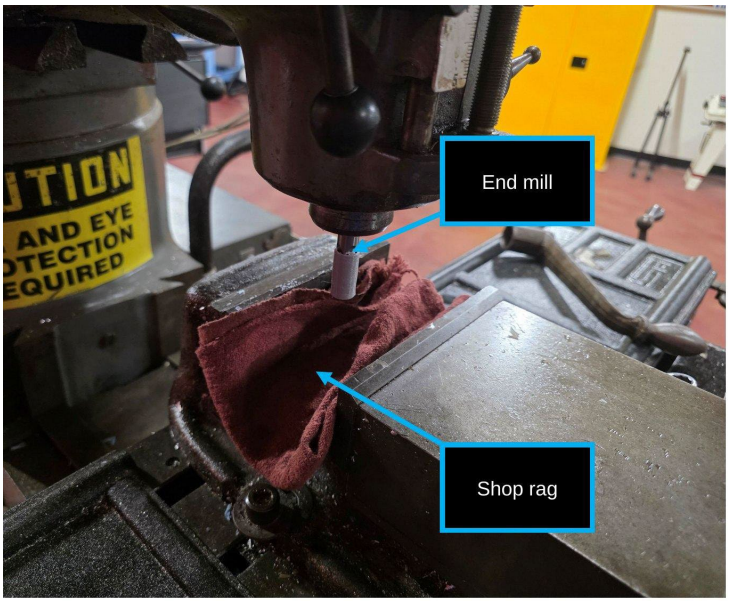


Figure 42. Shop rag placed under the end mill before loosening the drawbar.

3. Hold the **spindle brake lever** firmly (Figure 13) and use the **drawbar wrench** to loosen the **drawbar** counterclockwise (Figure 14). If you cannot reach the **spindle brake** or the **drawbar** while standing on the floor, ask your instructor for help. Do not stand on anything to reach them.

4. Once the **drawbar** is loose, put the **wrench** back on the **toolbox**. Hold the **end mill** by its **cover** with one hand and unscrew the **drawbar** the rest of the way by hand (Figure 12).

5. The **collet and end mill** should drop into your hand or onto the **rag** once the **drawbar** is free. If they stay stuck, lift the **drawbar** up a couple of inches and let it drop to knock the **collet** loose (Figure 43). If that does not work, stop and ask your instructor for help.



Figure 43. Freeing a stuck collet: lift the drawbar up a couple of inches and let it drop.

6. Take the **end mill** out of the **collet** by its **smooth end**. Leave the **protective cover** on.

7. Put all the tools back on top of the **toolbox** in the same places they were before you started: the **collet**, **end mill** with its **protective cover**, **drawbar wrench**, **vise handle**, **parallels**, **shop rag**, and **chip brush** (Figure 44). Check the **toolbox** against the picture before you leave.

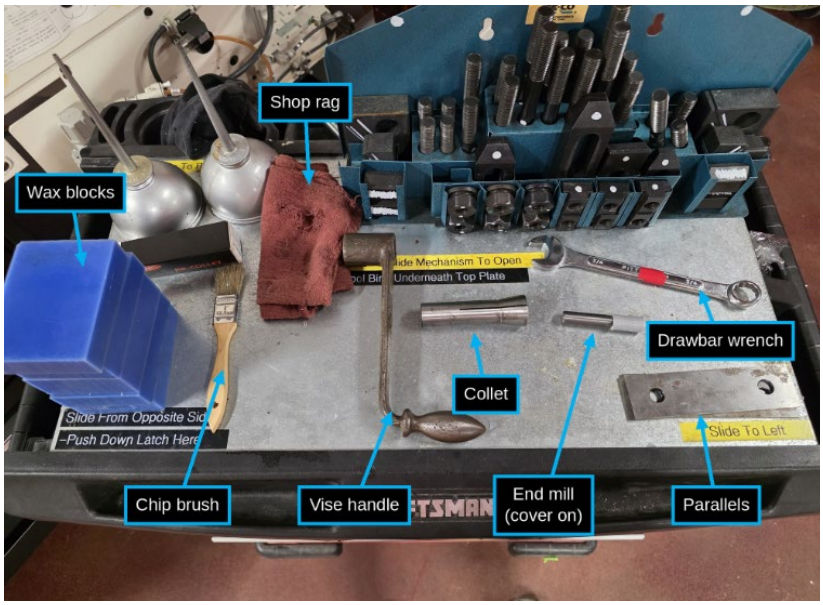


Figure 44. All tools back in their places on top of the toolbox.